NODE.JS

Part II

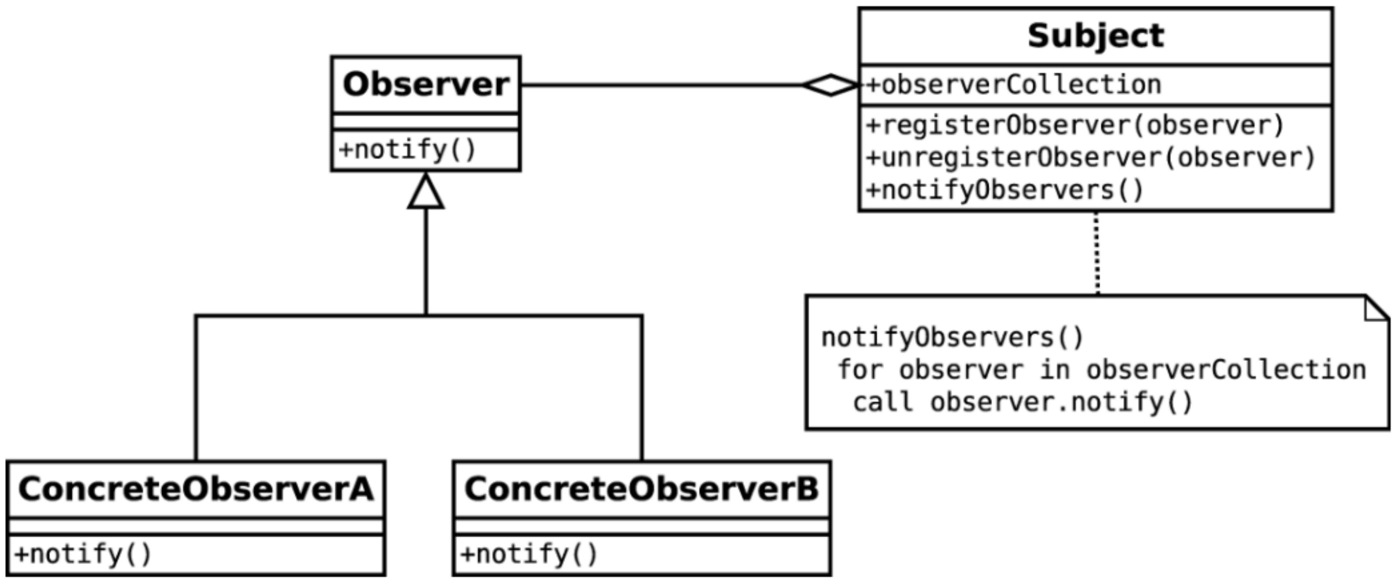
THE OBSERVER PATTERN

- Together with reactor, callbacks, and modules, this is one of the pillars of the platform and an absolute prerequisite for using many node-core and userland modules
- Definition: Pattern (observer): defines an object (called subject), which can notify a set of observers (or listeners), when a change in its state happens.
- The main difference from the callback pattern is that the subject can actually notify multiple observers, while a traditional continuation-passing style callback will usually propagate its result to only one listener, the callback
- No need to define interfaces as required in Java to implement this pattern

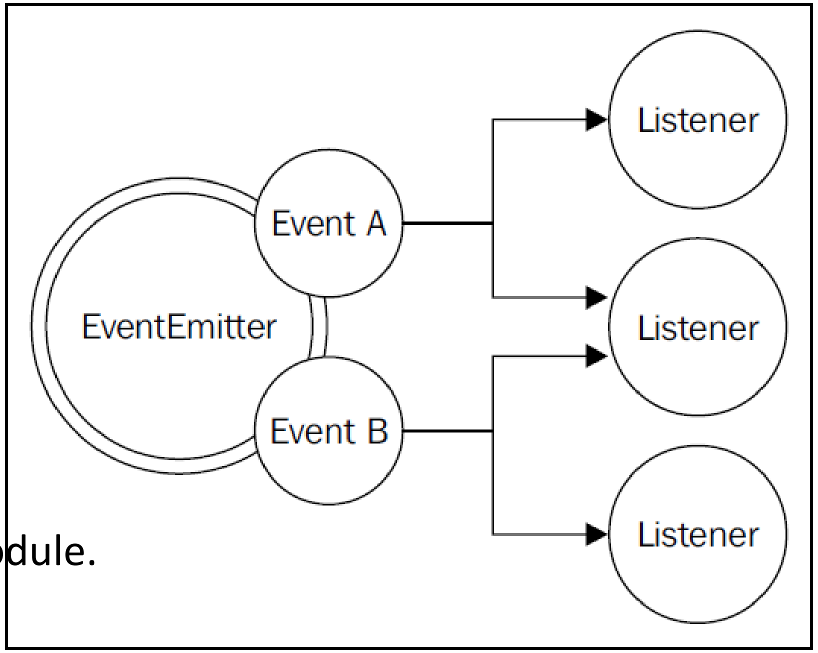
The observer pattern is already built into the core and is available through the **EventEmitter** class.

The `EventEmitter` class allows us to register one or more functions as listeners, which will be invoked when a particular event type is fired

THE OBSERVER PATTERN



EVENT EMITTER



- The `EventEmitter` is a prototype, and it is exported from the `events` core module.
- `Obtaining a reference of the EventEmitter`
  - `var EventEmitter = require('events').EventEmitter;`
  - `var eeInstance = new EventEmitter();`
- The essential methods of the `EventEmitter` are given as follows.
  - `on(event, listener)`: This method allows you to register a new listener (a function) for the given event type (a string)
  - `once(event, listener)`: This method registers a new listener, which is then removed after the event is emitted for the first time
  - `emit(event, [arg1], […])`: This method produces a new event and provides additional arguments to be passed to the listeners
  - `removeListener(event, listener)`: This method removes a listener for the specified event type

EVENT EMITTERS

- All the preceding methods will return the `EventEmitter` instance to allow chaining.
- There is a big difference between a listener and a traditional Node.js callback; in particular, the first argument is not an error, but it can be any data passed to emit() at the moment of its invocation.
- The `listener` function has the signature, `function([arg1], […])`, so it simply accepts the arguments provided the moment the event is emitted.
- Inside the listener, `this` refers to the instance of the `EventEmitter` that produces the event

EMITTING EVENTS

- The `EventEmitter` - as it happens for callbacks - cannot just throw exceptions when an error condition occurs, as they would be lost in the event loop if the event is emitted asynchronously.
- Instead, the convention is to emit a special event, called `error`, and to pass an `Error` object as an argument.
- A common dilemma when defining an asynchronous API is to check whether to use an `EventEmitter` or simply accept a callback.
- The general differentiating rule is semantic: callbacks should be used when a result must be returned in an asynchronous way; events should instead be used when there is a need to communicate that something has just happened

DESIGN STYLES TO COVER

- Reactor pattern
- Callbacks
- Modules
- Observer pattern

CLOSURE

```
function makeAdder(x) {
  return function (y) {
    return x + y;
  };
}
const add5 = makeAdder(5);
const add10 = makeAdder(10);

console.log(add5(2)); // 7
console.log(add10(2)); // 12
```

```
function makeFunc() {
  const name = 'Mozilla';
  function displayName() {
    console.log(name);
  }
  return displayName;
}

const myFunc = makeFunc();
myFunc();
```

- A **closure** is the combination of a function bundled together (enclosed) with references to its surrounding state (the **lexical environment**)
- When a function is returned from an outer function, it carries the environment with itself.
- **Lexical scoping**, which describes how a parser resolves variable names when functions are nested. The word lexical refers to the fact that lexical scoping uses the location where a variable is declared within the source code to determine where that variable is available.
- A closure gives you access to an outer function's scope from an inner function
- displayName() inner function is returned from the outer function before being executed

https://developer.mozilla.org/en-US/docs/Web/JavaScript/Closures#:~:text=A%20closure%20is%20the%20combination,scope%20from%20an%20inner%20function.

CALLBACKS-PROS AND CONS

- Closures and in-place definition of anonymous functions allow a smooth programming experience that doesn't require the developer to jump to other points in the code base.
- Sacrificing qualities such as modularity, reusability, and maintainability will sooner or later lead to the uncontrolled proliferation of callback nesting, the growth in the size of functions, and will lead to poor code organization.
- Most of the time, creating closures is not functionally needed, so it's more a matter of discipline than a problem related to asynchronous programming
- Recognizing that our code is becoming unwieldy—or even better, knowing in advance that it might become unwieldy—and then acting accordingly with the most adequate solution is what differentiates a novice from an expert.

- a command-line application that takes in a web URL as input and downloads its contents locally into a file
- The npm dependencies are
- `request`: A library to streamline HTTP calls
- `mkdirp`: A small utility to create directories recursively

```
function spider(url, callback) {
var filename = utilities.urlToFilename(url);
fs.exists(filename, function(exists) { //[1]
if(!exists) {
    console.log("Downloading " + url);
    request(url, function(err, response, body) {
    //[2]
    if(err) {
        callback(err);
    } else {
        mkdirp(path.dirname(filename),
        function(err) { //[3] if(err) {
                                  callback(err);
                                  } else {
                                      fs.writeFile(filename, body, function(err){
                                              if(err) {//[4]
                                          callback(err);
                                      } else {
                                          callback(null, filename, true);
                                      }
                                  });
                                  }
                                  });
                                  }
                                  });
                                  } else {
                                      callback(null, filename, false);
                                  }
                                  });
                              }
```

WEB-SPIDER APPLICATION

1. Checks if the URL was already downloaded by verifying that the corresponding file was not already created:

```
fs.exists(filecodename, function(exists) …
```

2. If the file is not found, the URL is downloaded using the following line of code:

```
request(url, function(err, response, body) …
```

3. Then, we make sure whether the directory that will contain the file exists or not:

```
mkdirp(path.dirname(filename), function(err) …
```

4. Finally, we write the body of the HTTP response to the filesystem:

```
fs.writeFile(filename, body, function(err) …
```

CALLBACK-HELL

- Even though the algorithm we implemented is really straightforward, the resulting code has several levels of indentation and is very hard to read.
- Implementing a similar function with direct style blocking API would be straightforward, and there would be very few chances to make it look so wrong
- The situation where the abundance of closures and in-place callback definitions transform the code into an unreadable and unmanageable blob is known as **callback hell**.
- It's one of the most well recognized and severe anti-patterns in Node.js and JavaScript in general.
- The typical structure of a code affected by this problem looks like

```
asyncFoo(function(err) {
asyncBar(function(err) {
asyncFooBar(function(err) {
[...]
});
});
});
```

CALLBACK-HELL

```
asyncFoo(function(err) {
    asyncBar(function(err) {
      asyncFooBar(function(err) {
            [...]
            });
        });
    });
```

- We can see how code written in this way assumes the shape of a pyramid due to the deep nesting and that's why it is also colloquially known as the *pyramid of doom*.
- The most evident problem with code such as the preceding one is the poor readability.
- Due to the nesting being too deep, it's almost impossible to keep track of where a function ends and where another one begins.
- Another issue is caused by the overlapping of the variable names used in each scope.
- When writing asynchronous code, the first rule to keep in mind is to not abuse closures when defining callbacks
- Most of the times, fixing the callback hell problem does not require any library, fancy technique, or change of paradigm but just some common sense

Hell to Heaven-How to Reach

These are some basic principles that can help us keep the nesting level low and improve the organization of our code in general:

- You must exit as soon as possible. Use return, continue, or break, depending on the context, to immediately exit the current statement instead of writing (and nesting) complete if/else statements. This will help keep our code shallow.
- You need to create named functions for callbacks, keeping them out of closures and passing intermediate results as arguments.
- Naming our functions will also make them look better in stack traces.
- You need to modularize the code. Split the code into smaller, reusable functions whenever it's possible.

```
if(err) {
callback(err);
} else {
//code to execute when there are no
errors
}
```

```
if(err) {
return callback(err);
}
//code to execute when there are no errors
```

- With this simple trick, we immediately have a reduction of the nesting level of our functions; it is easy and doesn't require any complex refactoring.
- We should never forget that the execution of our function will continue even after we invoke the callback.
- It is then important to insert a `return` instruction to block the execution of the rest of the function.
- Also note that it doesn't really matter what output is returned by the function; the real result (or error) is produced asynchronously and passed to the callback.
- The return value of the asynchronous function is usually ignored.
- This property allows us to write shortcuts such as the following:
- `return callback(...)`
- Instead of the slightly more verbose ones such as the following:
- `callback(...)`
- `return;`

PROMISES

- Promises are an abstraction that allow an asynchronous function to return an object called a **promise**, which represents the eventual result of the operation.
- In the promises jargon, we say that a promise is **pending** when the asynchronous operation is not yet complete, it's **fulfilled** when the operation successfully completes, and **rejected** when the operation terminates with an error.
- Once a promise is either fulfilled or rejected, it's considered **settled.**
- To receive the fulfillment value or the error (*reason*) associated with the rejection, we can use the `then()` method of the promise.
- The following is its signature:

```
promise.then([onFulfilled], [onRejected])
```

- Where `onFulfilled()` is a function that will eventually receive the fulfillment value of the promise, and `onRejected()` is another function that will receive the reason of the rejection (if any).
- Both functions are optional

PROMISE BASED API

```
asyncOperation(arg, function(err, result) {
    if(err) {
        //handle error
}
    //do stuff with result
    });
```

```
asyncOperation(arg)
.then(function(result) {
//do stuff with result
}, function(err) {
//handle error
});
```

- One crucial property of the `then()` method is that it synchronously returns another promise.
- If any of the `onFulfilled()` or `onRejected()` functions return a value `x`, the promise returned by the `then()` method will be as follows:
- • Fulfill with `x` if `x` is a value
- • Fulfill with the fulfillment value of `x` if `x` is a promise or a **thenable**
- • Reject with the eventual rejection reason of `x` if `x` is a promise or a thenable

PROMISE

- **.then()** is used to handle a fulfilled Promise and access its result.
- **.catch()** is used to handle a rejected Promise and catch any errors that may occur.
- **.finally()** is used to handle a settled Promise, regardless of whether the Promise resolved or rejected.

- A thenable is a promise-like object with a `then()` method.
- This term is used to indicate a promise that is *foreign* to the particular promise implementation in use.
- This feature allows us to build chains of promises, allowing easy aggregation and arrangement of asynchronous operations in several configurations.
- Also, if we don't specify an `onFulfilled()` or `onRejected()` handler, the fulfillment value or rejection reasons are automatically forwarded to the next promise in the chain

```
const myPromise = new Promise((resolve, reject) => {
  const success = true;
  if (success) {
    resolve('Operation was successful!');  } else {
    reject('Something went wrong.');
  }
});
myPromise
  .then(result => {
    console.log(result); // This will run if the Promise is fulfilled
  })
  .catch(error => {
    console.error(error); // This will run if the Promise is rejected
  })
  .finally(() => {
    console.log('The promise has completed'); // runs when the Promise is settled
  });
```

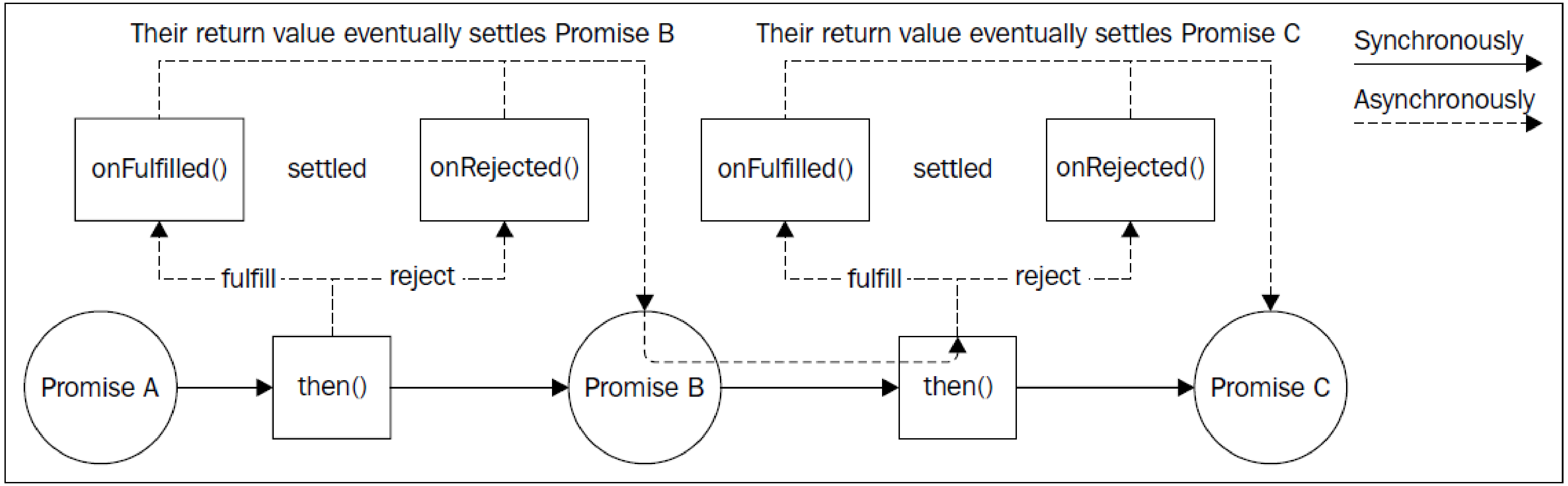
PROMISE CHAIN

- With a promise chain, sequential execution of tasks suddenly becomes a trivial operation
- When you chain Promises, each `.then()` block waits for the previous one to complete before it runs.

```
asyncOperation(arg)
.then(function(result1) {
    //returns another promise
    return asyncOperation(arg2);
    })
.then(function(result2) {
//returns a value
return 'done';
})
.then(undefined, function(err) {
//any error in the chain is caught
here
});
```

PROMISE CHAIN

- If an exception is thrown (using the `throw` statement) from the `onFulfilled()` or `onRejected()` handler, the promise returned by the `then()` method will automatically reject with the exception as the rejection reason.
- This is a tremendous advantage over CPS, as it means that with promises, exceptions will propagate automatically across the chain



MULTIPLE PROMISES

```
const promise1 = Promise.resolve(3);const
promise2 = 42;const promise3 = new
Promise((resolve, reject) => {
setTimeout(resolve, 100,
"foo");});Promise.all([promise1, promise2,
promise3]).then((values) => {
console.log(values);});
```

- Promise.all() returned promise fulfills when all of the input's promises fulfill (including when an empty iterable is passed), with an array of the fulfillment values.
- It rejects when any of the input's promises rejects, with this first rejection reason.
- Promise.allSettled() waits for all promises to either resolve or reject and returns an array of objects that describe the outcome of each Promise.
- Unlike **Promise.all()**, **Promise.allSettled()** does not short-circuit on failure. It waits for all promises to settle, even if some reject.
- This provides better error handling for batch operations, where you may want to know the status of all tasks, regardless of failure.

CONNECTING A DATABASE

- MongoDB is a NoSQL database used to store large amounts of data without any traditional relational database table.
- Instead of rows & columns, MongoDB used collections & documents to store data.
- A collection consists of a set of documents & a document consists of key-value pairs which are the basic unit of data in MongoDB
- Two approaches to connect a web app to a database

Using the databases' native query language, such as SQL.

Using an Object Relational Mapper ("ORM") or Object Document Mapper ("ODM").

CONNECTING DATABASE

- Object-Relational Mappers (or ORMs) can improve the developer experience customizing database interactions in the following ways:
- Abstracting away the need for query language.
- Managing serialization/deserialization of data into objects.
- Enforcing schema requirements.
- Because MongoDB is a non-relational database management system, ORMs are sometimes referred to as ODMs (Object Document Mappers)

CONNECTING DATABASE

- Several object-document mappers (ODMs) for Node.js are provided that let developers work with MongoDB data as objects.
- One popular ODM is **Mongoose**, which helps enforce a semi-rigid schema at the application level and provides features to assist with data modeling and manipulation.
- Mongoose is a third-party Node.js-based ODM library for MongoDB.
- The document storage and query system looks very much like JSON, and is hence familiar to JavaScript developers.
- It enforces a specific schema at the application layer – for a better structural representation (data types, error checking) at the web application level
- It offers a variety of hooks that is, pre and post middleware that allow developers to execute logic (such as hashing a password or logging data) before or after executing database operations like save or validate
- It also supports model validation, and other features.
- **Prisma**, another ORM/ODM, helps ensure data consistency by offering a type-safe database client.

CONNECTING TO A DATABASE

Step 1: Initialize npm on the directory and install the necessary modules. Also, create the index file

```
$ npm i express mongoose
```

Step 2: Initialise the express app and make it listen to a port on localhost.

```
const express = require("express");

const app = express();

app.listen(3000, () => console.log("Server is
running"));
```

CONNECTION

- To connect a Node.js application to MongoDB, we have to use a library called **Mongoose**
- `const mongoose = require("mongoose");`
- Connect method is invoked with URL and user credentials

```
mongoose.connect("mongodb://localhost:27017/newCollection", {
      useNewUrlParser: true,
      useUnifiedTopology: true
});
```

- createConnection() is utilized to create additional connection objects.
- It returns immediately; if you need to wait on the connection to be established you can call it with asPromise() to return a promise (mongoose.createConnection(mongoDB).asPromise()).

CONFIGURING THE DATABASE

```
const breakfastSchema = new Schema({
  eggs: {
    type: Number,
    min: [6, "Too few eggs"],
    max: 12,
    required: [true, "Why no eggs?"],
  },  drink: {
    type: String,
    enum: ["Coffee", "Tea", "Water"],
  },});
```

- A schema is defined
- A schema is a structure, that gives information about how the data is being stored in a collection.

```
const contactSchema = {
      email: String,
      query: String,
      };
```

- Then we have to create a model using that schema which is then used to store data in a document as objects
- `const Contact = mongoose.model("Contact", contactSchema);`

SCHEMA OBJECT

- When you call mongoose.model() on a schema, Mongoose compiles a model for you.
- The first argument is the singular name of the collection your model is for.
- Mongoose automatically looks for the plural, lowercased version of your model name.

```
Contact.create({ query: 'small' }, function (err, small) {
  if (err) return handleError(err);
  // saved!
});

// or, for inserting large batches of documents
Contact.insertMany([{ query: 'small' }], function(err) {
});
```

CRUD

Contact.find()

Contact.delete;

Contact.update();

```
async function run() {
// Create a new mongoose model
const personSchema = new mongoose.Schema({
name: String });
const Person = mongoose.model('Person',
personSchema);
// Create a change stream. The 'change'
event gets emitted when there's a // change
in the database
Person.watch(). on('change', data =>
console.log(new Date(), data));
```

STORING DATA

we are able to store data in our document

```
app.post("/contact", function (req, res) {
    console.log(req.body.email);
const contact = new Contact({
    email: req.body.email,
    query: req.body.query,
});
```

```
contact.save(function (err) {
    if (err) {
        throw err;
    } else {
        res.render("contact");
    }
});
```

MONGO DB CONNECTION

By using Mongoose we’re able to access the MongoDB database in an object-oriented way.

This means that we need to add a Mongoose schema

MongoDB's collections, by default, do not require their documents to have the same schema. That is:

- The documents in a single collection do not need to have the same set of fields and the data type for a field can differ across documents within a collection.
- To change the structure of the documents in a collection, such as add new fields, remove existing fields, or change the field values to a new type, update the documents to the new structure.

This flexibility facilitates the mapping of documents to an entity or an object. Each document can match the data fields of the represented entity, even if the document has substantial variation from other documents in the collection.

STORING FORM DATA WITH MULTIPLE FIELDS

```
app.post("/contact", function (req, res) {
    console.log(req.body.email);
const contact = new Contact({
    email: req.body.email,
    query: req.body.query,
});

contact.save(function (err) {
    if (err) {
        throw err;
    } else {
        res.render("contact");
    }
});
```

```
contact = new Contact(req.body);
```

STORING FORM DATA WITH MULTIPLE FIELDS

```
let contact = new Contact(req.body);
contact.save()
.then(contact => {
            res.status(200).json({'contact': 'contact added successfully'});
        })
        .catch(err => {
            res.status(400).send('adding new contact failed');
        });
```